ELT

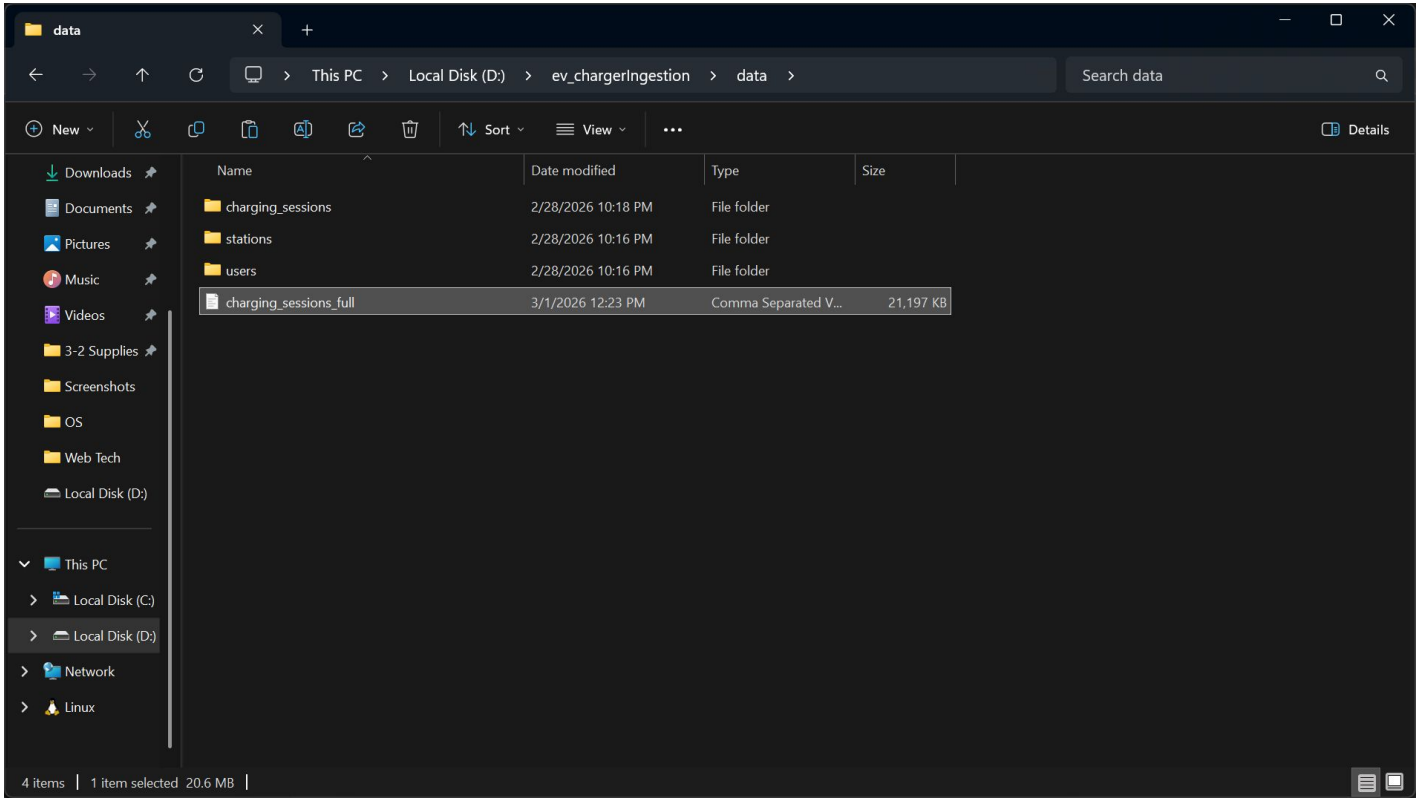
Data Ingestion into Apache Druid using Python Script

Title: Data Ingestion Workflow

Steps

1. Raw CSV dataset is stored locally.
2. Python script prepares an ingestion specification.
3. Python sends ingestion request to Apache Druid API.
4. Druid creates an indexing task.
5. Data is ingested and stored as a datasource.
6. Users run analytical SQL queries on the ingested data.

Dataset Preparation



Python Script for Ingestion

File Edit Selection View ··· druid_ingestion

EXPLORER

DRUID_INGESTION

.venv

ingest_druid.py

Welcome | ingest_druid.py

ingest_druid.py > ...

```python
import requests

url = "http://localhost:8081/druid/indexer/v1/task"

spec = {
  "type": "index_parallel",
  "spec": {
    "dataSchema": {
      "dataSource": "ev_charging_two",
      "timestampSpec": {
        "column": "start_time",
        "format": "yyyy-MM-dd HH:mm:ss"
      },
      "dimensionsSpec": {
        "useSchemaDiscovery": True
      }
    },
    "ioConfig": {
      "type": "index_parallel",
      "inputSource": {
        "type": "local",
        "baseDir": "/mnt/d/ev_chargerIngestion/data",
        "filter": "charging_sessions_full1.csv"
      },
      "inputFormat": {
        "type": "csv",
        "findColumnsFromHeader": True
      }
    },
    "tuningConfig": {
      "type": "index_parallel"
    }
  }
}

response = requests.post(url, json=spec)

print(response.text)
```

OUTLINE

TIMELINE

0 0 | Ln 18, Col 18 | Spaces: 2 | UTF-8 | CRLF | { } Python | 3.12.10 (.venv) | .venv (3.12.10) | Go Live

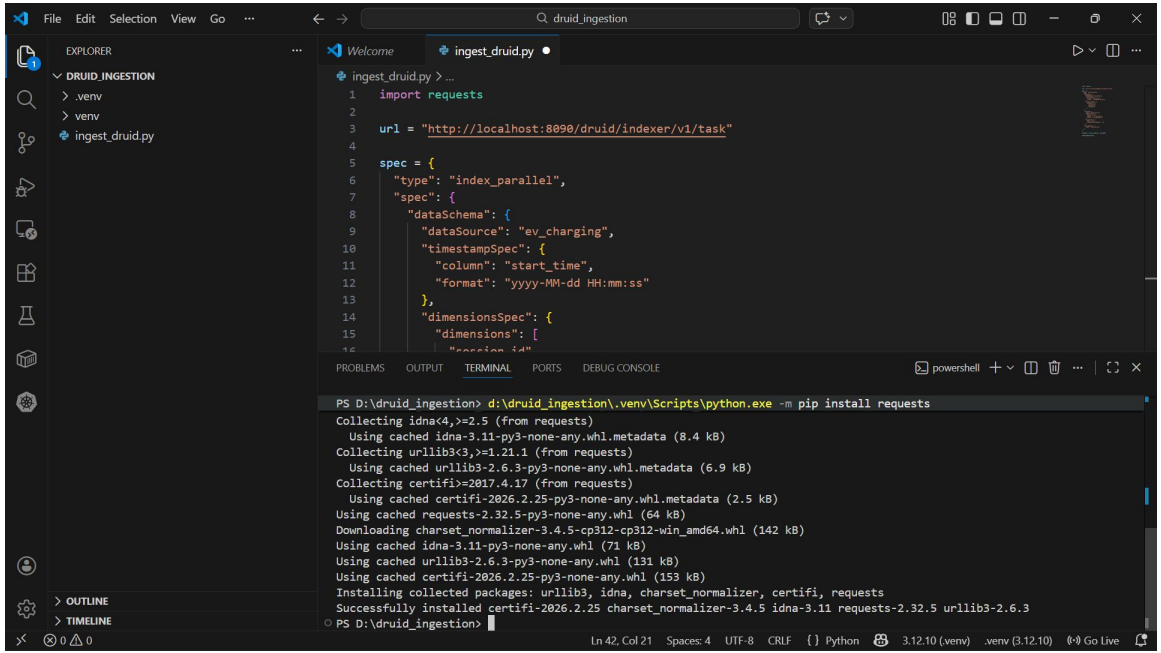
Installing Python Dependencies

Title: Installing Required Libraries

→Before running the ingestion script, required Python libraries must be installed.

Command

pip install requests



Running the Python Script

File Edit Selection View ... druid_ingestion

EXPLORER

DRUID_INGESTION

.venv

ingest_druid.py

Welcome | ingest_druid.py

ingest_druid.py > ...

```python
url = "http://localhost:8081/druid/indexer/v1/task"

spec = {
  "type": "index_parallel",
  "spec": {
    "dataSchema": {
      "dataSource": "ev_charging_two",
      "timestampSpec": {
        "column": "start_time",
        "format": "yyyy-MM-dd HH:mm:ss"
      },
      "dimensionsSpec": {
        "useSchemaDiscovery": True
      }
    },
    "ioConfig": {
      "type": "index_parallel",
      "inputSource": {
        "type": "local",
        "baseDir": "/mnt/d/ev_chargerIngestion/data",
        "filter": "charging_sessions_full1.csv"
      },
      "inputFormat": {
        "type": "csv",
```

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE

powershell

```
PS D:\druid_ingestion> & d:\druid_ingestion\.venv\Scripts\Activate.ps1
(.venv) PS D:\druid_ingestion> python ingest_druid.py
{"task":"index_parallel_ev_charging_edacbblb_2026-03-16T04:55:12.006Z"}
(.venv) PS D:\druid_ingestion> python ingest_druid.py
{"task":"index_parallel_ev_charging_two_cjgmollo_2026-03-16T04:56:32.664Z"}
(.venv) PS D:\druid_ingestion> python ingest_druid.py
{"task":"index_parallel_ev_charging_two_mjhanbdi_2026-03-16T05:06:28.740Z"}
(.venv) PS D:\druid_ingestion>
```

OUTLINE

TIMELINE

0 0 | Ln 18, Col 18 | Spaces: 2 | UTF-8 | CRLF | { } Python | 3.12.10 (.venv) | .venv (3.12.10) | Go Live

Apache Druid

Apache Druid is a high-performance, real-time analytics database designed for fast aggregation queries on large datasets.It is optimized for time-series data and interactive analytics.

-Druid can process millions to billions of records quickly, making it ideal for analytics dashboards.

-It uses columnar storage, indexing, and distributed architecture for high performance.

<u>Key points:</u>

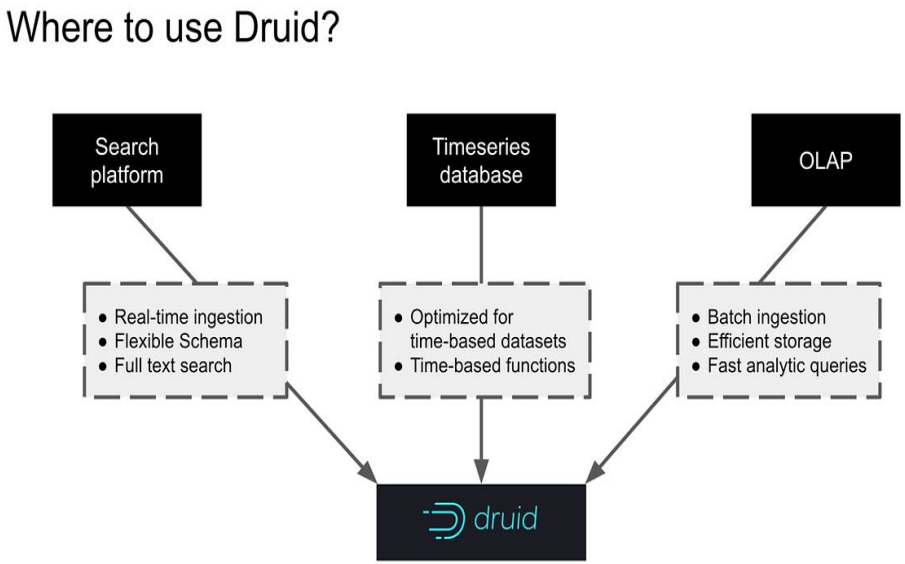
- Fast aggregation queries.
- Real-time data ingestion and analysis.
- Optimized for time-series data.
- Pre-aggregation using rollup to reduce storage and improve speed.
- Integration with BI tools like Superset, Tableau, and Power BI.
- Column-oriented storage for high query performance.

<u>Key Points (From ELT Perspective)</u>:

❖ Load First, Transform Later:
  ➢ Druid allows loading raw data and performing transformations using SQL queries after ingestion.

- Real-Time ELT Processing
  - Data can be ingested and transformed almost immediately for analytics.
- Supports SQL-based Transformations
  - Druid SQL allows applying calculations, filtering, grouping, and derived columns.
- Real-Time ELT Processing
  - Data can be ingested and transformed almost immediately for analytics.
- Rollup for Pre-Aggregation
  - Data can be aggregated during ingestion to speed up analytical queries.

Where Apache Druid is Used :



Companies Using Apache Druid :

Apache Druid is widely used by many leading technology companies for real-time analytics and large-scale data processing.

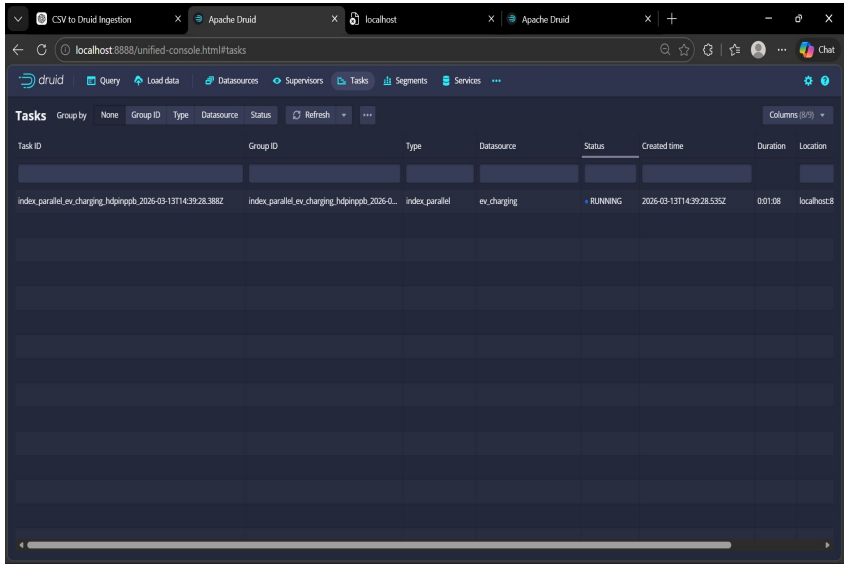
Major Companies Using Apache Druid:

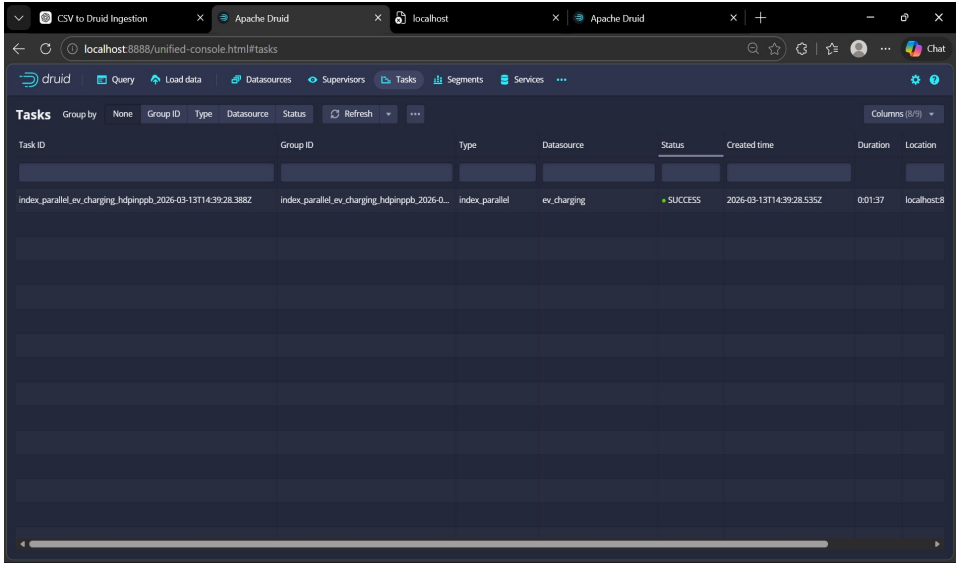
- Netflix – analyzes user streaming behavior and real-time viewing analytics.
- eBay – processes large-scale marketplace data for analytics and monitoring.
- Airbnb – performs real-time analytics on booking and user activity data.
- Twitter – analyzes social media events and engagement metrics in real time.
- Lyft – monitors ride data, driver activity, and operational analytics.
- Pinterest – analyzes user engagement and recommendation analytics.

Start Apache Druid

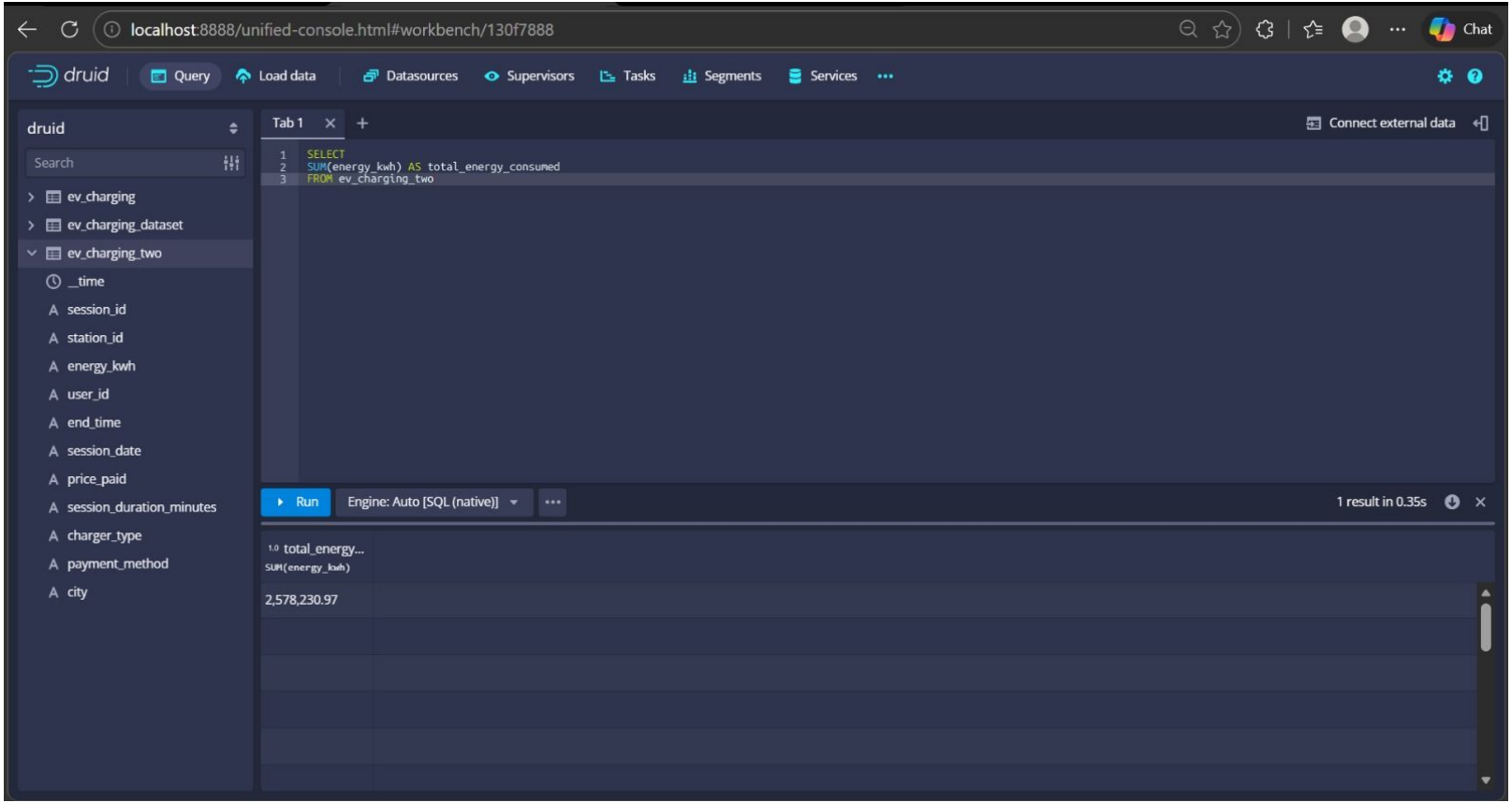
```
abraruddin1@Zeuss:~$ cd apache-druid-36.0.0
abraruddin1@Zeuss:~/apache-druid-36.0.0$ bin/start-micro-quickstart
[Mon Mar 16 04:34:52 2026] Starting Apache Druid.
[Mon Mar 16 04:34:52 2026] Open http://localhost:8888/ in your browser to access the web console.
[Mon Mar 16 04:34:52 2026] Or, if you have enabled TLS, use https on port 9088.
[Mon Mar 16 04:34:52 2026] Starting services with log directory [/home/abraruddin1/apache-druid-36.0.0/log].
[Mon Mar 16 04:34:52 2026] Running command[zk]: bin/run-zk conf
[Mon Mar 16 04:34:52 2026] Running command[coordinator-overlord]: bin/run-druid coordinator-overlord conf/druid/single-s
erver/micro-quickstart
[Mon Mar 16 04:34:52 2026] Running command[broker]: bin/run-druid broker conf/druid/single-server/micro-quickstart
[Mon Mar 16 04:34:52 2026] Running command[router]: bin/run-druid router conf/druid/single-server/micro-quickstart
[Mon Mar 16 04:34:52 2026] Running command[historical]: bin/run-druid historical conf/druid/single-server/micro-quicksta
rt
[Mon Mar 16 04:34:52 2026] Running command[middleManager]: bin/run-druid middleManager conf/druid/single-server/micro-qu
ickstart
```

Monitoring Ingestion Tasks



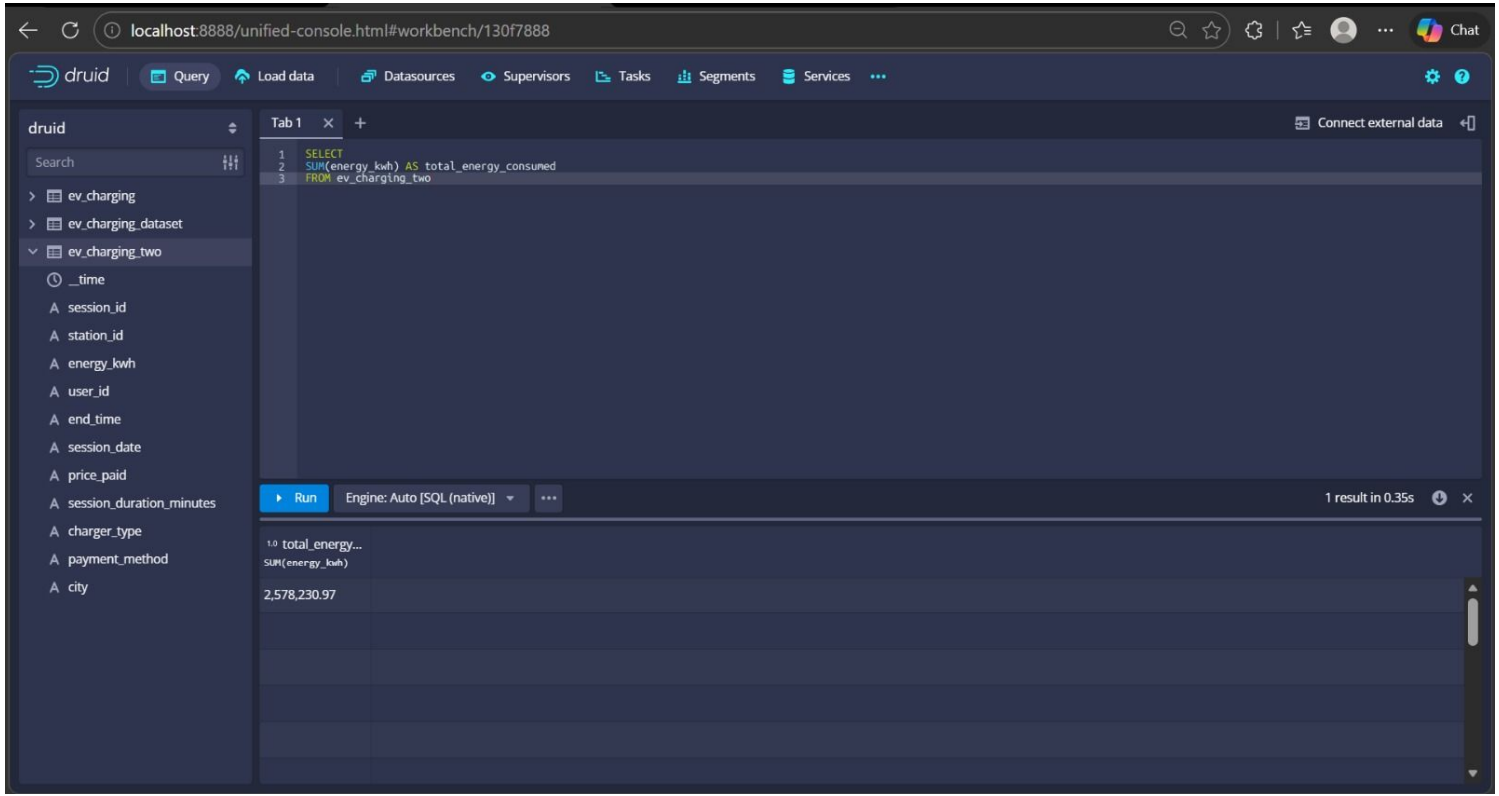


Querying the Ingested Data



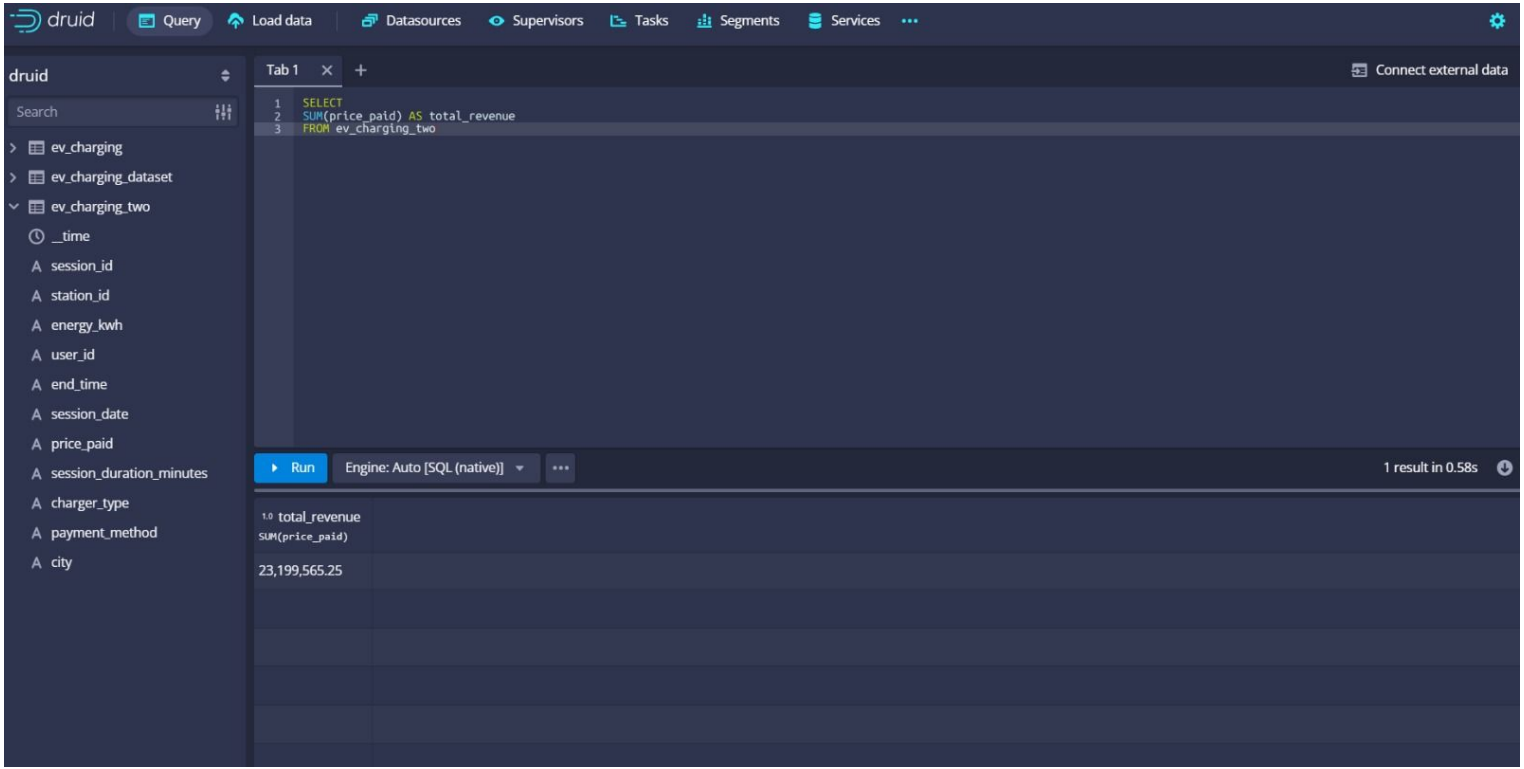
Data Transformation

How much electricity was delivered?



Data Transformation

Total Revenue Generated?



Data Transformation

Daily Revenue

druid | Query | Load data | Datasources | Supervisors | Tasks | Segments | Services

druid

Search

- ev_charging
- ev_charging_dataset
- ev_charging_two
  - __time
  - session_id
  - station_id
  - energy_kwh
  - user_id
  - end_time
  - session_date
  - price_paid
  - session_duration_minutes
  - charger_type
  - payment_method
  - city

Tab 1 | Tab 2 | Tab 3 | Connect

```
SELECT
session_date,
SUM(price_paid) AS total_revenue
FROM ev_charging_two
GROUP BY session_date
ORDER BY session_date
```

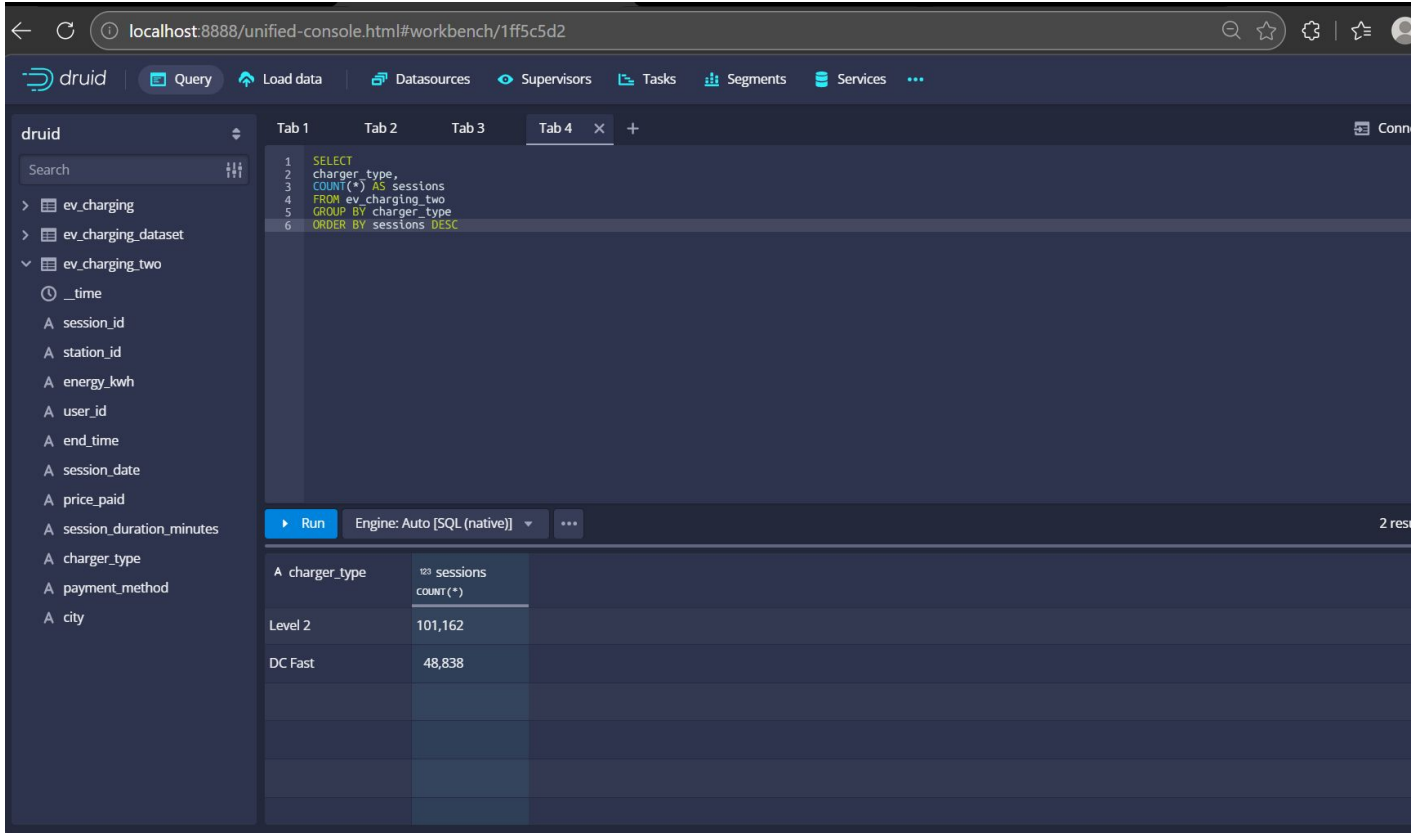
Run | Engine: Auto [SQL (native)] | 30 results

session_date	total_revenue SUM(price_paid)
2026-02-01	768,329.9
2026-02-02	770,982.71
2026-02-03	779,849.73
2026-02-04	779,991.13
2026-02-05	773,233.81

Showing 1-20 of 30

Data Transformation

Payment Method Distribution



Data Transformation

Total Revenue Generated?

druid | Query | Load data | Datasources | Supervisors | Tasks | Segments | Services

druid

Search

- ev_charging
- ev_charging_dataset
- ev_charging_two
  - __time
  - session_id
  - station_id
  - energy_kwh
  - user_id
  - end_time
  - session_date
  - price_paid
  - session_duration_minutes
  - charger_type
  - payment_method
  - city

Tab 1 | Connect external data

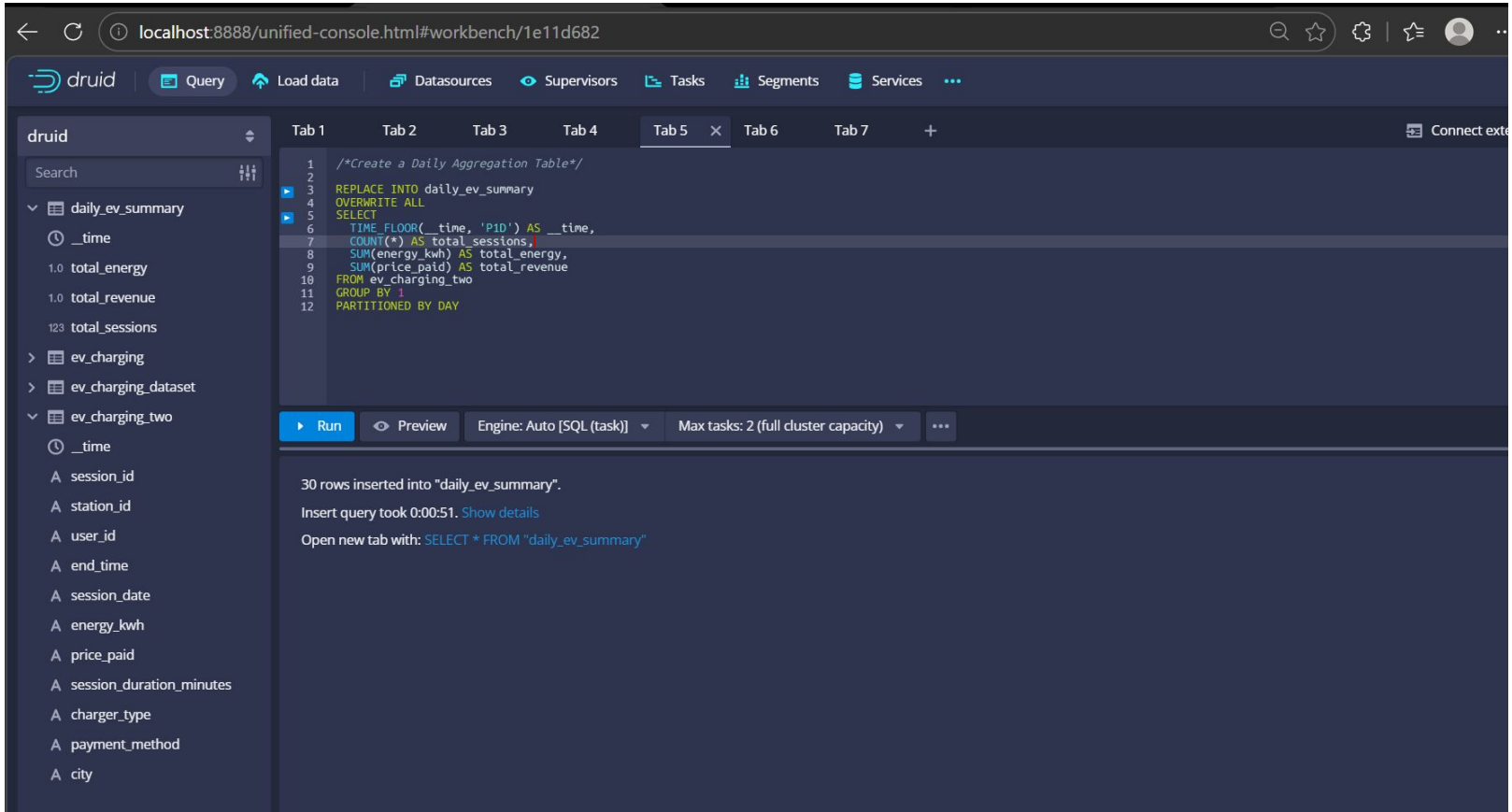
```
SELECT
SUM(price_paid) AS total_revenue
FROM ev_charging_two
```

Run | Engine: Auto [SQL (native)] | 1 result in 0.58s

total_revenue SUM(price_paid)
23,199,565.25

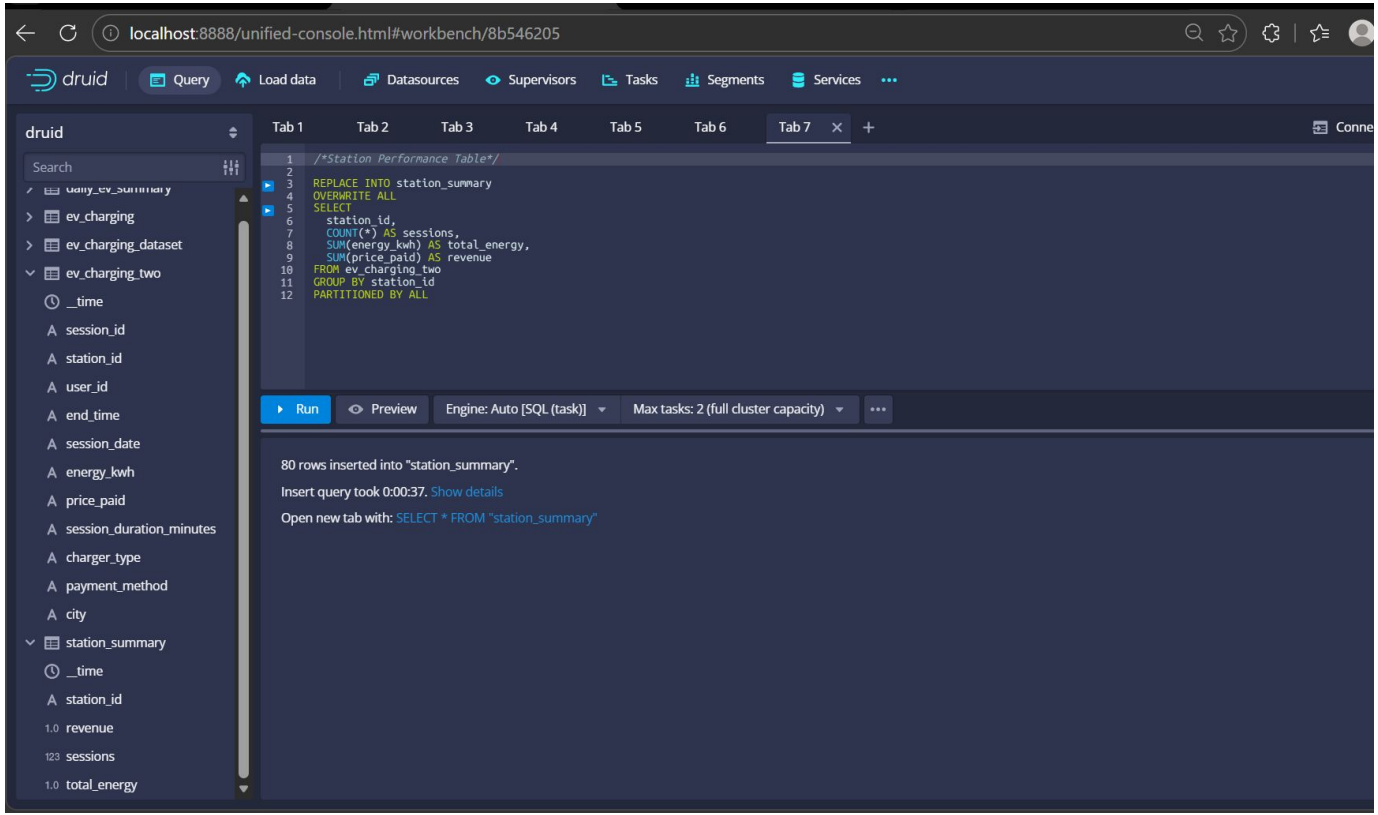
Data Transformation

Create a Daily Aggregation Table



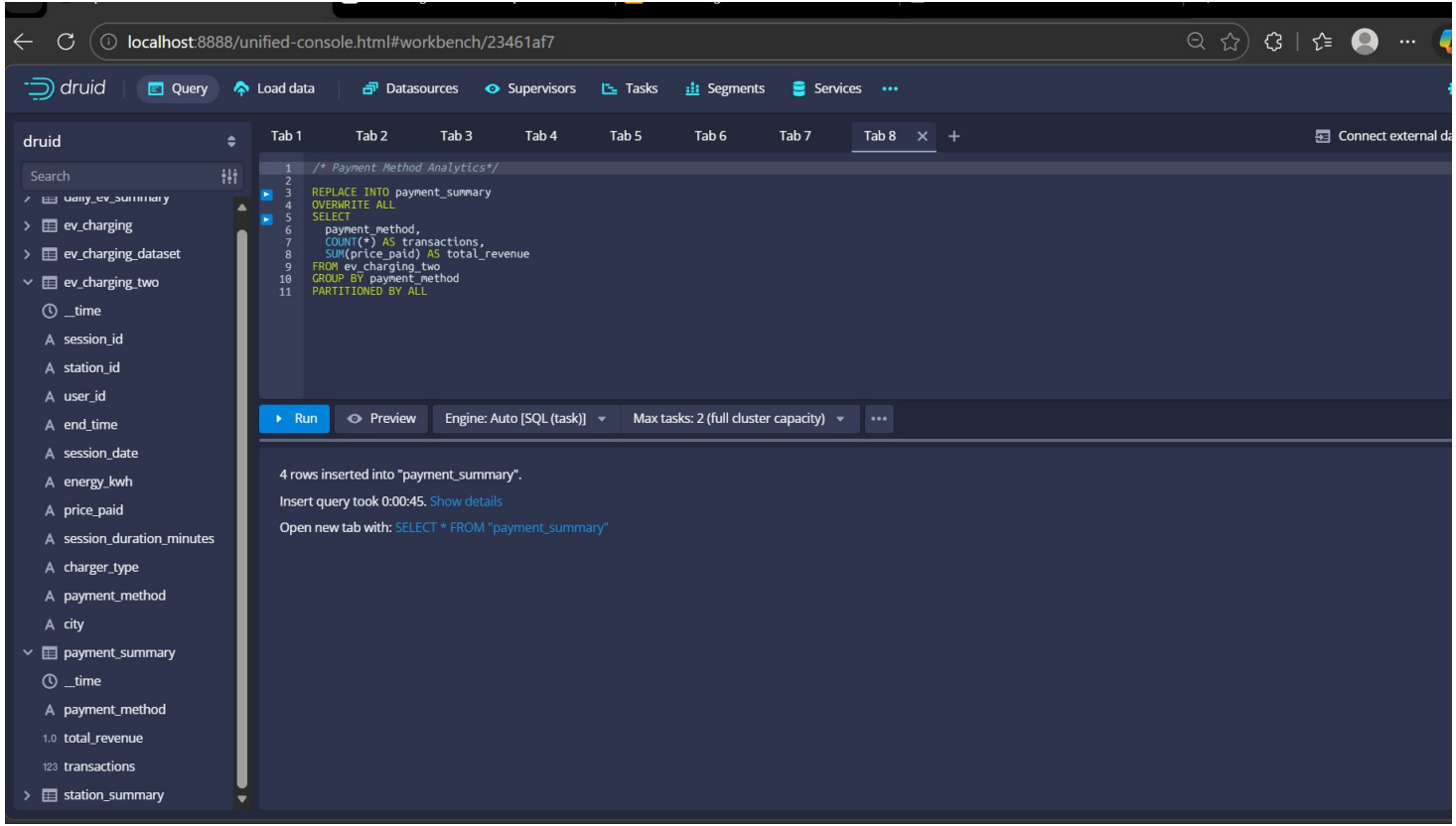
Data Transformation

Station Performance Table



Data Transformation

Payment Method Analytics

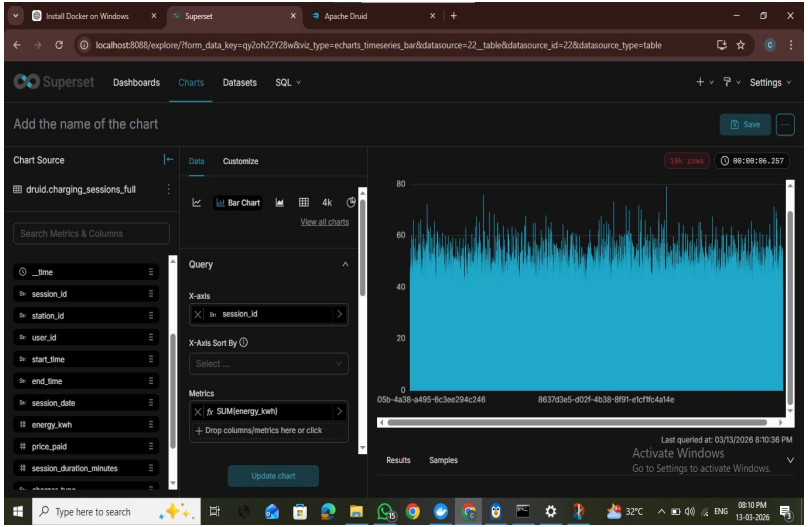


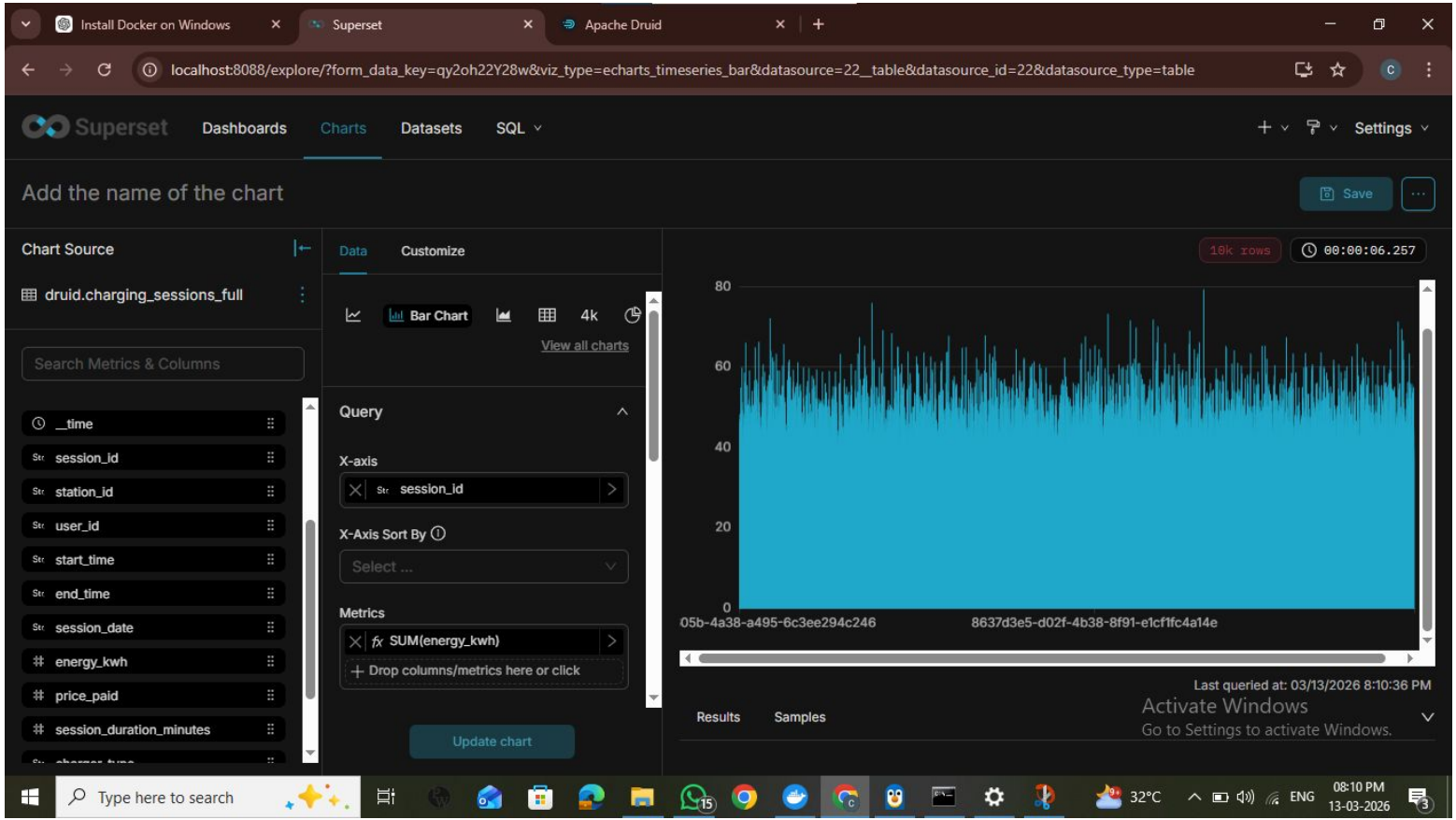
❖ Apache Superset(visualization)

Superset connects to databases and allows users to query data and visualize it as charts, graphs, and dashboards without writing complex programs.

Why It Is Used

1. **Data Visualization**
   Converts raw data into charts, graphs, and dashboards.
2. **Business Intelligence (BI)**
   Helps analyze data trends and insights for decision making.
3. **SQL Querying**
   Users can run SQL queries directly on connected databases.
4. **Interactive Dashboards**
   Multiple charts can be combined into a single dashboard for monitoring data.
5. **Connects to Many Databases**
   It can connect to systems like:
   - PostgreSQL(Neon)
   - Apache Druid
   - MySQL, Snowflake, etc.




Install Docker on Windows
Superset
Apache Druid
localhost:8088/explore/?form_data_key=qy2oh22Y28w&viz_type=echarts_timeseries_bar&datasource=22__table&datasource_id=22&datasource_type=table
Superset
Dashboards
Charts
Datasets
SQL
Settings
Add the name of the chart
Save
Chart Source
druid.charging_sessions_full
Search Metrics & Columns
__time
session_id
station_id
user_id
start_time
end_time
session_date
energy_kwh
price_paid
session_duration_minutes
Data
Customize
Bar Chart
4k
View all charts
Query
X-axis
session_id
X-Axis Sort By
Select ...
Metrics
SUM(energy_kwh)
Drop columns/metrics here or click
Update chart
10k rows
00:00:06.257
80
60
40
20
0
05b-4a38-a495-6c3ee294c246
8637d3e5-d02f-4b38-8f91-e1cf1fc4a14e
Last queried at: 03/13/2026 8:10:36 PM
Results
Samples
Activate Windows
Go to Settings to activate Windows.
Type here to search
32°C
ENG
08:10 PM
13-03-2026

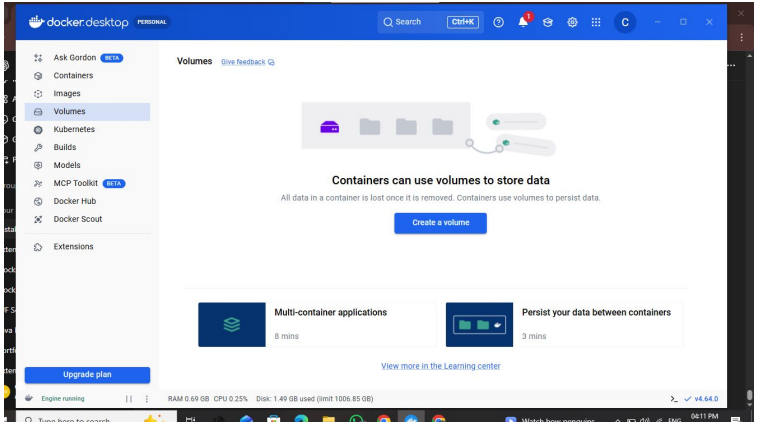
❖ Docker

Docker is a tool used to run applications inside **containers.**A container includes the **application + all its required libraries + dependencies** in one package.This makes the application run the **same way on any system.**

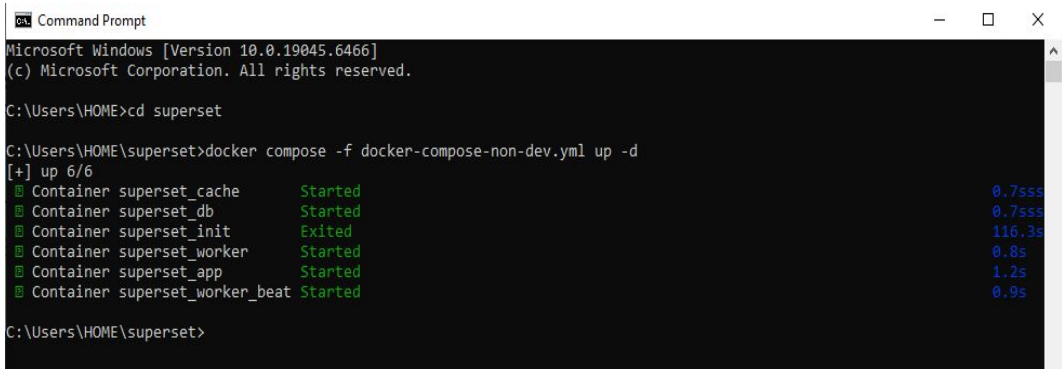
Docker helps in :

- **Easy Installation**
- **Avoid Dependency Problems**
- **Same Environment Everywhere**
- **Easy Deployment**

So with the help of docker we downloaded the required dependency and setup the environment to run the superset in the system .



Steps to run superset :



1. Open cmd run the command.

   Enter these commands :

   cd superset.

   docker compose -f docker-compose-non-dev.yml up -d.

2. Open http://localhost:8088.
3. Enter your username.
4. Enter your password.
5. You are ready to go.

Superset

Settings Login

Sign in

Enter your login and password below:

Username:

chandu

Password:

Sign in

Activate Windows
Go to Settings to activate Windows.

Superset Dashboards Charts Datasets SQL Settings

Home

Recents

Dashboards

Favorite Mine All + Dashboard View All »

Video Game Sales
Modified 4 minutes ago Published

FCC New Coder Survey...
Modified 4 minutes ago Published

Sales Dashboard
Modified 4 minutes ago Published

Charts

Favorite Mine All + Chart View All »

Top 10 Games: Proporti...
Modified 4 minutes ago

Most Dominant Platforms
Modified 4 minutes ago

Games per Genre
Modified 4 minutes ago

Activate Windows
Go to Settings to activate Windows.